

YU ZHANG

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EDUCATIONAL BACKGROUND

University of Galway	Galway, Ireland	Sept 2025 - Sept 2029(expected)
• Ph.D in Computer Science		
University of Galway	Galway, Ireland	Sept 2024 - Sept 2025
• M.Sc in Computer Science - Artificial Intelligence		
Heilongjiang University of Science and Technology	Harbin, China	Oct 2020 - Jun 2024
• B.Eng in Computer Science and Technology (Major Degree), GPA: 3.55 (85.5%)		
• B.Sc in Mathematics and Applied Mathematics (Minor Degree), GPA: 3.75 (87.5%)		

RESEARCH INTERESTS

Reinforcement Learning, Optimisation, Evolutionary Computing

RESEARCH & PROJECT EXPERIENCE

Advanced Genetic Algorithm Enhancements for Complex Optimization Problems	Galway, Ireland
• Designed and implemented Dynamic Mutation Rate and Simulated Annealing Acceptance Criterion to enhance the exploration-exploitation balance and improve the algorithm's ability to escape local optima, significantly increasing global search efficiency.	
• Introduced a Restart Mechanism Near the Best Solution, focusing search efforts around high-quality solutions while maintaining population diversity, effectively addressing stagnation in optimization tasks.	
• Applied an Elitism Strategy and enhanced genetic diversity with Two-Point Crossover, preserving top-performing individuals and improving the convergence and robustness of solutions.	
• Tested the improved GA on multiple complex optimization scenarios, achieving superior performance in terms of convergence speed, solution quality, and adaptability, showcasing its practicality for real-world problems.	
Research on Machine Learning-Based Clustering for Cardiac Health Assessment	Galway, Ireland
• Conducted comprehensive data preprocessing to ensure effective clustering, including handling missing values through median imputation for numerical variables and mode imputation for categorical variables to maintain data consistency. Outliers were identified and removed using Z-scores (>3), followed by standardization to align feature scales and normalization to enhance clustering performance.	
• Applied t-SNE for dimensionality reduction to analyze the influence of each variable on clustering. By sequentially removing variables and visualizing their impact, 9 critical features were identified as most relevant for clustering analysis.	
• Implemented K-means++ and hierarchical clustering algorithms to process the dataset. K-means++ ensured optimal cluster initialization, producing well-distributed clusters. Both methods achieved identical results, with a Silhouette Score of 0.86 and a Davies-Bouldin Score of 0.17, indicating high-quality and well-separated clusters.	
• Evaluated clustering performance using metrics such as Silhouette Score and Davies-Bouldin Score. K-means++ with $k=4$ produced compact, well-separated clusters, supported by the dendrogram from hierarchical clustering, validating the preprocessing and clustering approach.	
Research on Machine Learning and Evolutionary Computing for Perinatal Risk Prediction	Galway, Ireland
• Conducted comprehensive data preprocessing for a dataset with 1,014 entries, including outlier removal using Z-Scores and normalization for neural network models.	
• Applied Multi-Layer Perceptron (MLP) neural networks and Random Forest algorithms to classify perinatal risks into three categories: low, medium, and high.	
• Developed a genetic algorithm (evolutionary computing approach) to optimize Random Forest hyperparameters, improving model accuracy from 84% to 88.27%.	
• Evaluated model performance using advanced metrics (precision, recall, F1-score), achieving balanced classification results across all categories.	
• Explored ethical implications of combining machine learning and evolutionary computing in healthcare, emphasizing potential biases and the need for data privacy.	
Research on Second-hand Car Price Evaluation Based on K-means and Random Forest Algorithm	Harbin, China
Graduation Capstone Project for B. Sc in Mathematics and Applied Mathematics, Score: 87.14/100	
• Conducted comprehensive preprocessing of a dataset containing 30,000 second-hand cars, including outlier removal, missing value imputation, and feature engineering; identified critical variables highly correlated with car prices using correlation matrix heatmaps to enhance model input quality.	
• Applied PCA for dimensionality reduction and designed a K-means clustering algorithm to group vehicles into three categories, optimizing clusters with the Elbow Method and validating with a Silhouette Coefficient of 0.75.	
• Developed and optimized a random forest model and genetic algorithm to predict car prices by selecting optimal hyperparameters and improving model performance.	
• Optimized model parameters and assessed performance using advanced metrics such as Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), R-squared (R^2), and Mean Absolute Error (MAE) to ensure accuracy, robustness, and generalizability.	

- Applied the model to predict prices for 10 unknown second-hand cars, showcasing real-world application and providing valuable pricing insights for dealers and buyers.

Design of Crack Detection System for USTH Road Based on Image Processing

Harbin, China

Graduation Capstone Project for B. Eng in Computer Science and Technology, Score: 93/100 (rank: 1/158)

- Developed a Python-based road crack image collection system, enabling automated image capture both inside and outside the Heilongjiang University of Science and Technology campus. The system efficiently manages image naming, local storage, and scalability for larger datasets.
- Designed an image processing system in MatLab to process and display collected images, incorporating techniques such as histogram equalization and binary filtering. The system also features automatic crack severity classification and assigns repair tasks to relevant personnel.
- Built a comprehensive web-based platform using Vue.js and SpringBoot, supporting CRUD operations for road crack data. The platform enhances data management, streamlines administrative tasks, and provides a user-friendly interface for maintenance teams.

Research on Optimized Autonomous Parking Systems

Harbin, China

Team Leader, Second Prize in the MathorCup National College Mathematical Modeling Challenge

- Optimized the kinematic model of low-speed vehicle motion by improving the relationship between steering angles and minimum turning radius to meet the demands of complex parking scenarios.
- Applied the A* algorithm to enhance parking path planning, generating efficient and smooth parking trajectories while satisfying path constraints.
- Utilized the entropy weighting method to comprehensively evaluate parking space selection strategies, optimizing multi-factor weights such as travel distance, viewing angle, and parking space distribution for optimal parking choices.
- Conducted MATLAB-based simulation to optimize dynamic parking trajectory modeling, improving computational efficiency and adaptability to dynamic environments.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

Heilongjiang University of Science and Technology AI Association

Sep 2021 - Sep 2022

Technical Consultant, Training Group Leader

- **Mentorship:** Guided and mentored a team of 60+ junior members, focusing on practical skills in machine learning, data analysis, and model optimization.
- **Project Oversight:** Provided technical oversight for team-led AI projects, ensuring high-quality outcomes through code reviews, algorithm optimization, and project presentations.
- **Community Building:** Fostered a collaborative learning environment by creating a knowledge-sharing platform, encouraging members to discuss challenges and share innovative solutions.

AWARDS & DISTINCTION

- Excellent Capstone Project (**Rank: 1/158**), Awarded by Heilongjiang University of Science and Technology Jun,2024
- **Second Prize**, The Chinese Mathematics Competition (CMC), Awarded by the China Mathematical Society Mar 2023
- **Second Prize**, National English Competition for College Students (NECCS), Awarded by the International English Language Teachers Association Oct 2022
- **Second Prize**, MathorCup National College Mathematical Modeling Challenge Competition, Awarded by the Chinese Society for the Promotion of Science and Economics Mathematics Research Oct 2022
- **Third Prize**, Lanqiao Cup National Software and Information Technology Professionals Competition, Awarded by the Ministry of Industry and Information Technology Talent Exchange Center May 2022
- **Third Prize**, MathorCup National College Mathematical Modeling Challenge Competition - Big Data Competition, Awarded by the Chinese Society for the Promotion of Science and Economics Mathematics Research Mar 2022
- **Third Prize**, Northeast Three Provinces Mathematical Modeling League, Awarded by the Northeast Three Provinces Mathematical Modeling League Organizing Committee Aug 2021

SKILLS & LANGUAGE

PROGRAMMING SKILLS: Python (Advanced), C (Advanced), Java (Intermediate)

PYTHON FRAMEWORKS: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, SciPy

PROFESSIONAL APPLICATIONS: MatLab, LINGO, SPSS, CoppeliaSim

LANGUAGE: English (Proficient, IELTS 6.5); Mandarin (Native)

